

Ann Von Holle

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Population Health Scientist

Doctoral-level epidemiologist with a strong quantitative background demonstrating skills in designing, analyzing, interpreting, and scientific writing for epidemiological research studies. Proven record of accomplishment in peer-reviewed journals. Collaborated in healthcare payer space to develop real world evidence-based strategies supporting health outcomes.

Core competencies

epidemiologic methods, biostatistics, real-world evidence (RWE), scientific writing, predictive modeling, unsupervised machine learning, clustering, observational studies, data visualization, R programming

Research Experience

American Medical Group Association, Remote Senior Population Health Research Analyst

February 2025 – Present

Scope: Responsible for leading research projects within the AMGA Research and Analytics group, including ideation through publication and dissemination.

- Lead research projects including developing study concepts, designing research protocols, conducting research, and disseminating findings through presentations and peer reviewed manuscripts.
- Design and conduct analyses using real-world data and advanced epidemiological and analytic methods as part of a collaborative process.
- Apply advanced analytic skills to design statistically sound analytical approaches and compelling data visualizations supporting population health improvement for medical groups and integrated health care systems across the US.
- Collaborate with multiple stakeholders including funders, members of the programming team, and other analysts to ensure projects are completed successfully.

Optum Genomics, Remote Associate Research Director

April 2023 - July 2024

Scope: Conducted real world evidence (RWE) studies and decision analytic modeling to evaluate health and economic outcomes, focusing on genetic diagnostic test validity and cancer detection. Contributed to white papers and literature reviews on molecular diagnostics.

- Planned, designed, and executed a simulation study on carrier screening to evaluate health and economic outcomes, leveraging decision analytic models and R software.
- Conducted decision analytic modeling using R software, incorporating clinical validity measures, simulations, and claims data to support evidence-based conclusions for carrier screening.
- Contributed to research focusing on cancer early detection, evaluating the analytical and clinical validity of molecular diagnostic tests by analyzing medical claims data with R software.
- Collaborated on the development of white papers addressing the analytical and clinical validity of genetic and molecular lab-developed tests, providing expert insights and recommendations.
- Conducted comprehensive literature reviews on the quality of molecular diagnostic tests, integrating findings to inform research and practice.

National Institute of Environmental Health Sciences, Research Triangle Park, NC**Special Volunteer****April 2023 - present**

Complete manuscript submission process for research completed during my postdoctoral fellowship.

Postdoctoral Fellow**July 2018 - April 2023****Biostatistics and Computational Biology Branch, Advisor: Clarice R. Weinberg**

Scope: Conducted observational research studies by developing research concepts, specifying statistical analysis plans, doing formal analyses, and authoring manuscripts related to breast cancer outcomes and serum iron biomarkers. Used advanced analytical techniques and high-performance cluster computing, communicated findings at national conferences, and mentored students.

- Developed research concepts and hypotheses, planned eight lead-author manuscripts, and selected appropriate data and analytic methods as part of a collaborative process.
- Applied advanced analytic approaches such as time-to-event analysis, longitudinal mixed-effects models, machine learning, meta-analyses, and data visualization in research on breast cancer outcomes and biomarker exposures.
- Authored initial drafts and reviewed subsequent drafts for epidemiological research manuscripts
- Presented research findings internally and at national conferences.
- Performed large-scale data analyses using a high-performance computing environment
- Mentored graduate and undergraduate students in developing analysis plans, conducting research, and presenting results as part of NIH training programs.

University of North Carolina, Chapel Hill, Chapel Hill, NC**Graduate Researcher****August 2013 - June 2018****Department of Epidemiology, Advisor: Kari E. North**

Scope: As a doctoral candidate in a genetic epidemiology training program, used latent growth mixed effects modeling to assess infant anthropometric growth and its association with adolescent lipid levels in relation to sociodemographic factors and genetic risk variants. Authored three lead manuscripts, secured a predoctoral grant from the American Heart Association, and used reproducible research techniques with R, Markdown, LaTeX, and Git.

- Using mixed effects models and latent growth modeling, characterized infant anthropometric growth as it relates to sociodemographic risk factors, its impact on lipid levels in adolescence, and its role as an effect modifier in the association between genetic risk variants and lipid levels. Developed scientific hypotheses, created analysis plans, analyzed observational cohort data, and wrote manuscripts resulting in three lead author manuscripts.
- Employed advanced statistical techniques such as latent growth mixed effects modeling, factor analysis, and non-linear random effects models to estimate patterns of infant growth and their association with adolescent lipid levels
- Developed and submitted a pre-doctoral American Heart Association grant application titled "Infant Growth Trajectories and Dyslipidemia in Adolescence," funded in 2016 (Award Number 16PRE29200008).
- Implemented reproducible research techniques by documenting writing and statistical analyses in R, Markdown, and LaTeX, and maintained version control with Git.

Biostatistician, Chapel Hill, NC**September 2005 - September 2014****Department of Psychiatry, Center of Excellence for Eating Disorders (CEED)**

Scope: Worked across various academic levels and collaborated with researchers to plan, conduct, and interpret statistical analyses for peer-reviewed manuscripts

- Collaborated in the planning and interpretation of statistical analyses for research by faculty, post-doctoral fellows, and students.
- Analyzed observational, clinical trial, and medical record data using regression and structural equation models in SAS and R, contributing to over forty peer-reviewed publications.
- Authored statistical methods sections, created data visualizations, and ensured data integrity for peer-reviewed publications.

Education

Doctor of Philosophy (Ph.D.), Epidemiology, University of North Carolina at Chapel Hill, Chapel Hill, NC | **Thesis:** Infant Growth Trajectories and Lipid Levels in Adolescence

Master of Science (M.S.), Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC

Bachelor of Arts (B.A.), Applied Mathematics, University of California, San Diego, San Diego, CA

Recent Publications

*Indicates a mentored student's work. †Indicates equal authorship

1. **Sedentary Time and Breast Cancer Risk in the Sister Study Cohort.**
Donzella, Sidney M., Rogers M, O'Brien KM, **Von Holle A**, Sandler DP, Weinberg CR, and Díaz Santana MV. (2025); Cancer Epidemiology, Biomarkers & Prevention 34(11):1955–62.
doi:10.1158/1055-9965.EPI-25-0510
2. **Association Between Body Iron Status and Biological Aging.**
Von Holle A, Ramamurthy S*, Díaz Santana MV, et al. *Nutrients*. (2025);17(9):1409.
doi:[10.3390/nu17091409](https://doi.org/10.3390/nu17091409)
3. **Association between two common SNPs, rs6564851 and rs6420424, and lutein and zeaxanthin levels in a cohort of U.S. women**
Von Holle A†, Shi M, O'Brien KM, Sandler DP, Weinberg CR, Park YM† *Frontiers in Nutrition* (2024): 11. doi: 10.3389/fnut.2024.1372393
4. **Assessment of iron status**
Von Holle A. *Current Opinion in Clinical Nutrition and Metabolic Care*. (2024);27(5):397-401.
doi:10.1097/MCO.0000000000001050
5. **BMI and Breast Cancer Risk around Age at Menopause**
Von Holle A, Adami HO, Baglietto L, et al. *Cancer Epidemiology*. 89 (2024):102545, DOI: 10.1016/j.canep.2024.102545
6. **Seasonal Patterns in Trace Elements Assessed in Toenails**
Wojcik, Kaitlyn M.*†, **Von Holle, Ann**†, O'Brien KM, White AJ, Karagas MR, Levine KE, Jackson BP, and Weinberg CR. *Environmental Advances*. 15 (2024) :100496, DOI: 10.1016/j.envadv.2024.100496
7. **Toenail and Serum Levels as Biomarkers of Iron Status in Pre-and Postmenopausal Women: Correlations and Stability over Eight-Year Follow-Up**
Von Holle, Ann, O'Brien KM, Sandler DP, Janicek R, Karagas MR, White AJ, Niehoff NM, Levine KE, Jackson BP, and Weinberg CR. *Scientific Reports*. 14 (2024) :1682, DOI: 10.1038/s41598-023-50506-5
8. **Associations between Serum Iron Biomarkers and Breast Cancer Tumor Size**
Von Holle, Ann, Thompson, Rachel L.*, O'Brien KM, Sandler DP, and Weinberg CR. *Cancer Research Communications*. (2023), DOI: 10.1158/2767-9764.CRC-23-0205
9. **Development and Validation of a Prediction Model for Iron Status in a Large U.S. Cohort of Women**
Von Holle, Ann, O'Brien KM, Janicek R, and Weinberg CR. *Scientific Reports*. 13 (2023) :17309, DOI: 10.1038/s41598-023-42993-3
10. **Infant Growth Trajectories and Lipid Levels in Adolescence: Evidence From a Chilean Infancy Cohort**
Von Holle A, North KE, Gahagan S, Blanco E, Burrows R, Lozoff B, Howard AG, Justice AE, Graff M, and Voruganti S. *American Journal of Epidemiology*. (2022) :kwac057, DOI: 10.1093/aje/kwac057

11. **Does This Adult Patient Have Hypertension?: The Rational Clinical Examination Systematic Review.**
Viera AJ, Yano Y, Lin FC, Simel DL, Yun J, Dave G, **Von Holle A**, Viera LA, Shimbo D, Hardy ST, Donahue KE, Hinderliter A, Voisin CE, and Jonas DE.
JAMA. 326 (2021) :339–47, DOI: 10.1001/jama.2021.4533
12. **Evidence for familial clustering in breast cancer age of onset.**
Von Holle A, O'Brien KM, Sandler DP, and Weinberg CR.
International Journal of Epidemiology. 50 (2021) :97–104, DOI: 10.1093/ije/dyaa201
13. **Association Between Serum Iron Biomarkers and Breast Cancer.**
Von Holle A, O'Brien KM, Sandler DP, Janicek R, and Weinberg CR.
Cancer Epidemiology, Biomarkers & Prevention. 30 (2021) :422–5, DOI: 10.1158/1055-9965.EPI-20-0715
14. **The perils of standardizing infant weight to assess weight change differences across exposure groups.**
Von Holle, A, North, K. E., Tao, R, and Gahagan, S (2018). Annals of Epidemiology, 28(8), 515–520. <https://doi.org/10.1016/j.annepidem.2018.04.006>

Full publication list available at

<https://www.ncbi.nlm.nih.gov/myncbi/ann.von%20holle.1/bibliography/public/>

Publications In Review or In Preparation

1. (In press) Primary Adherence to medications for the treatment of major depressive disorder: a descriptive study using real-world data.
Von Holle, A, Mohl, JT, Zuckerman, RB, Obenrader, J, Ciemens, EL (2026). Journal of Managed Care + Specialty Pharmacy.
2. (In review) Inflammatory Marker Profiles Vary by BMI Across Crohn's Disease, Ulcerative Colitis, Psoriasis, and Psoriatic Arthritis.
Shields, S, **Von Holle, A**, Mohl, JT, Zuckerman, RB, Ciemens, EL (2026). Frontiers in Immunology.
3. (In preparation) Associations between male-origin microchimerism and breast, ovarian, and uterine cancer. (2026)
Von Holle, Ann, O'Brien KM, Sandler DP, Weinberg CR

Grants and awards

Grants

Individual Pre-Doctoral Fellowship Award (Award: 16PRE29200008)

American Heart Association (Total: \$51,900)

2016 - 2018

Awards

Intramural Paper of the Month

Environmental Factor, NIEHS Newsletter

May 2021

Title: Evidence for familial clustering in breast cancer age of onset.

Oral presentations

*Represents trainee's work for which Ann Von Holle was a mentor.

Association between Body Iron Status and Biological Aging, Durham, NC

2023

Ramamurthy S*, Von Holle A, Diáz-Santana, Kresovich J, Taylor J, Xu Z, O'brien Km, Sandler Dp, Weinberg Cr

National Institute of Environmental Health Sciences Scholars Connect Spring Symposium.

Infant Growth Trajectories and Lipid Levels in Adolescence: Evidence from a Chilean Infancy Cohort, New Orleans, LA

Von Holle A, North Ke, Gahagan S, Blanco, E, ET AL. American Heart Association EPI/Lifestyle Sessions

2018

Poster presentations

*Represents trainee's work for which Ann Von Holle was a mentor.

Associations between Serum Iron Biomarkers and Breast Cancer Tumor Size, Orlando, FL 2023
VON HOLLE A, THOMPSON RL*, O'BRIEN KM, SANDLER DP, WEINBERG CR American Association of Cancer Research Annual Meeting

Association between Body Iron Status and the Aging Process, Virtual 2022
 RAMAMURTHY S*, **VON HOLLE A**, DIÁZ-SANTANA, KRESOVICH J, TAYLOR J, XU Z, O'BRIEN KM, SANDLER DP, WEINBERG CR
 National Institute of Environmental Health Sciences Scholars Connect Poster Sessions

Seasonal Patterns in Toenail Metals, Virtual 2022
 WOJCIK KM*, **VON HOLLE A**, O'BRIEN KM, SANDLER DP, WHITE AJ, WEINBERG CR
 National Institute of Environmental Health Sciences Summer Internship Program Poster Sessions

Characterizing the association between BMI and breast cancer risk by menopause status, Chicago, IL 2022
 And age-time **VON HOLLE A**, O'BRIEN KM, SANDLER DP, SCHOEMAKER M, SWERDLOW A, NICHOLS HB, JONES ME, JANICEK R, THE PREMENOPAUSAL BREAST CANCER COLLABORATIVE GROUP, WEINBERG CR
 Society for Epidemiologic Research

Associations between Serum Iron Biomarkers and Breast Cancer Tumor Size, Virtual 2021
 THOMPSON RL, **VON HOLLE A**, O'BRIEN KM, SANDLER DP, WEINBERG CR
 National Institute of Environmental Health Sciences Summer Internship Program Poster Sessions

Predictive models for iron status in postmenopausal women, Virtual 2021
VON HOLLE A, O'BRIEN KM, SANDLER DP, JANICEK R, WEINBERG CR
 Society for Epidemiologic Research

Toenail and serum measures as biomarkers of iron levels, Virtual 2021
VON HOLLE A, O'BRIEN KM, SANDLER DP, JANICEK R, KARAGAS MR, JACKSON BP, WHITE, AJ, WEINBERG CR
 Society for Epidemiologic Research

Association Between Serum Iron Biomarkers and Breast Cancer., Virtual 2020
VON HOLLE A, O'BRIEN KM, SANDLER DP, JANICEK R, WEINBERG CR
 Society for Epidemiologic Research

Familial aggregation of age of onset for breast cancer risk, Minneapolis, MN 2019
VON HOLLE A, O'BRIEN K, SANDLER DP, WEINBERG CR
 Society for Epidemiologic Research

Sociodemographic predictors of early postnatal growth, Baltimore, MD 2018
VON HOLLE A, NORTH KE, GAHAGAN S, BLANCO E, ET AL.
 Society for Epidemiologic Research

Infant Growth as An Effect Modifier Of Genetic-Lipid Associations: Evidence From A Chilean, New Orleans, LA Infancy Cohort
VON HOLLE A, NORTH KE, GAHAGAN S, BLANCO E, ET AL.
 American Heart Association EPI/Lifestyle 2018

Confirmatory factor analysis (CFA) of multivariate lipid distributions across time and race-ethnic groups: United States, 2003-2012, Portland, OR
VON HOLLE A, NORTH KE
 American Heart Association EPI/Lifestyle 2017

Lipid-related Genetic Variants and Lipid Outcomes in a Cohort of Chilean Children, Portland, OR
VON HOLLE A, JUSTICE, A, NORTH KE, ET AL.
 American Heart Association EPI/Lifestyle 2017

The utility of standardized or crude weight measures in modeling of postnatal growth trajectories: Are there differences? , Miami, FL
VON HOLLE A, NORTH KE, GAHAGAN S, TAO R
 Epidemiology Congress of the Americas 2016

Childhood BMI Associated with Low HDL-C Levels in Adolescence in a Chilean Cohort, Baltimore, MD
VON HOLLE A, NORTH KE, KANG E, BLANCO E, GAHAGAN S
 American Heart Association EPI/Lifestyle 2015

Teaching and Mentoring Experience

National Institute of Environmental Health Sciences, Research Triangle Park, NC

- Mentor, 1 undergraduate student, NIEHS Scholars Connect Program 2022 - 2023
- Mentor, 2 graduate students, Summer Internship Program 2021, 2022
- Guest lecturer, "Introduction to Statistics and Experimental Design" January 2020
- Co-lead, "Mentoring of Junior Faculty and Fellows" session, Biostatistics and Computational Biology Branch retreat October 2019

University of North Carolina, Chapel Hill, Chapel Hill, NC

- Teaching Assistant, Advanced Epidemiologic Methods (<https://github.com/UNC-EPID-722/2016>) Spring 2016

Related Professional Experience

Academic and Professional Service

Journal Peer Review

BMJ Open | International Journal of Cancer | Cancer Epidemiology, Biomarkers & Prevention | BMC Public Health | Epidemiology | BMC Pregnancy and Childbirth | Journal of Eating Disorders | Nutrition and Metabolism | Cancer Medicine | Clinical Epigenetics

Society for Epidemiologic Research

- Annual meeting abstract reviewer 2021 - 2022
- Symposium co-chair: "Polygenic Risk Scores: challenges and opportunities for population health equity" December 2020

National Institute of Environmental Health Sciences, Research Triangle Park, NC

- Special volunteer: Publish manuscripts based on research completed during my post-doctoral fellowship. April 2024 - Present
- Trainee assembly steering committee member 2020 - 2022
- Summer intern program poster judge 2019 - 2021
- NIH Annual Graduate Student Research Symposium poster judge 2022
- Internal manuscript peer reviewer 2018 - 2022

Professional Development Training**National Institute of Environmental Health Sciences, Research Triangle Park, NC**

- Becoming a resilient scientist, 5-week course September - November 2022
- Research Mentor Training, 4-week course May 2019, May 2021
- 'Writing from the Reader's Perspective' 1-day workshop November 2019
- 'Write Winning NIH Grant Proposals' 1-day workshop February 2019
- 'Scientists Teaching Science', 9-week pedagogy course Spring 2019

Professional Associations

Society for Epidemiologic Research | American Association for Cancer Research